

17 D Using Rayleigh's Criterion

$$\theta_{\min} = \frac{\lambda}{a}$$

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If angular separation of light sources θ is greater than $\theta_{\min}$, the two images will be resolved (distinguished on the screen).

To reduce $\theta_{\min}$, one can reduce λ or increase a so options A and C will help to distinguish the two images.

For option C, by reducing distance D , the actual angular separation of the light sources θ gets larger so it is larger than $\theta_{\min}$. This change also helps to improve the ability to distinguish the two images.

For option D, changing L has no impact on θ or $\theta_{\min}$ so it is the correct answer.